Hugo Marquerie 24/01/2026

Prop convergencia uniforme continuidad uniforme

Proposición 1. *Sean X, Y espacios métricos, $f\colon X \to Y$ y sea $(f_n)_{n\in\mathbb{N}}$ una sucesión de funciones $f_n\colon X \to Y$ uniformemente continuas que converge uniformemente a f*

$$\implies f \text{ es uniformemente continua.}$$

***Demostración*:** Sea $\varepsilon > 0$. Como $f_n \underset{n\to\infty}{\rightrightarrows} f$, $\exists N \in \mathbb{N} : \forall x \in X : d_Y(f_N(x), f(x)) < \frac{\varepsilon}{3}$.

Entonces, como f_N es uniformemente continua, $\exists \delta > 0 : \forall x_1, x_2 \in X : d_X(x_1, x_2) < \delta \implies d_Y(f_N(x_1), f_N(x_2)) < \frac{\varepsilon}{3}$. Por tanto, si $d_X(x_1, x_2) < \delta$, por la desigualdad triangular,

$$\begin{aligned} d_Y(f(x_1), f(x_2)) &\leqslant d_Y(f(x_1), f_N(x_1)) + d_Y(f_N(x_1), f_N(x_2)) + d_Y(f_N(x_2), f(x_2)) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Luego f es uniformemente continua. ■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.